

Math 103

Class 1

1. **Statistics**: is the science of conducting studies to collect, organize, summarize, analyze and show conclusions from data.
2. **Students study statistics for several reasons:**
 - a. Students must be able to read and understand the various statistical studies performed in their field. To have this understanding, they must be knowledgeable about the vocabulary, symbols, concepts, and statistical procedures used in these studies.
 - b. Students may be called on to conduct research in their field, since statistical procedures are basic to all research. To accomplish this; they must be able to design experiments, collect, organize, analyze, and summarize data, and possibly make reliable predictions or forecasts for future use. They must also be able to communicate the results of the study in their own words.
 - c. Students can also use the knowledge gained from studying statistics to become better consumers and citizens. For example they can make intelligent decisions about what products to purchase based on consumer studies, government spending based on utilization studies, and so on.
3. **A variable**: is a characteristic or attribute that can assume different values.

Data: are the values (measurements or observations) that the variables can assume.

Random variables: are those whose values can be determined by Chance.
4. **Descriptive Statistics**: consists of the collection, organization, summation and presentation of data.
5. **Inferential Statistics**: consists of generalizing from samples to populations, performing hypothesis testing, determining relationships among variables and making predictions.

Population: all subjects that are being studied.

Sample: a subgroup of the population.

Hypothesis testing: making decisions by evaluating claims about A population - based on information attained from samples.

6. Types of variables

a. Qualitative: Can be put into distinct categories.

Examples: Gender
Religious preference
Geographic locations

b. Quantitative: Numerical in nature and can be ordered or ranked.

Examples: Age
Height
Weight
Body temperature

aa. Discrete variables assume values that can be counted.

Examples: Number of students enrolled at JUST
Number of people in Irbid
Number of grade complaints received
by the registrar's office

bb. Continuous variables can assume all values between any two specific values. They are obtained by measuring.

Examples: Temperature
Height
Weight

Rounding 200 lb. To the nearest pound
The boundaries are 199.5 - 200.5 lb.

The boundaries of a continuous variable are given in one Additional decimal place and always end with the digit 5.

<u>Measurement</u>	<u>Boundaries</u>
18 inches	17.5 - 18.5 inches
25.20° C	25.15 - 25.25° C
46.48 hours	46.475 - 46.485 hours
19.523 mg	19.5225 - 19.5235 mg

7. **Measurement Scales:**

How are the variables categorized, counted or measured?

a. Nominal level of measurement

No overlapping (mutually exclusive)
No order
No ranking

Examples: Zip Code
Gender
Eye color
Political affiliation
Religious affiliation
Major Field
Nationality

b. Ordinal level of measurement

Data can be ranked
Precise differences in ranks do not exist.

Examples: A, B, C, D, F grades
1st place, 2nd place
Good, fair, poor
Small, medium, large

c. Interval level of measurement

Precise differences do exist

Examples: SAT scores
I.Q.
Temperature

d. Ratio level of measurement

Same characteristics as Interval - but - there is
A true zero and objects can be compared by ratios.

Examples: Height
Weight
Time
Salary
Age

8. Collecting Data

a. Telephone Surveys

Advantage: People are candid.

Disadvantage: People are not home.

b. Mailed Surveys

Advantage: Less expensive.

Disadvantage: Low number of responses.

c. Personal Interview

Advantage: In depth responses

Disadvantage: Costly

d. To get unbiased samples, 4 basic methods are used.

aa. Random Sampling

Uses random numbers

Number each subject in the population.

bb. Systematic Sampling

Number each subject; then select every k^{th} number.

Select first one at random.

cc. Stratified Sampling

Divide population into groups (strata) according to some characteristic. Then pick samples from each group.

Frosh Soph Jr. Sr.

dd. Cluster Sampling

Uses intact groups that are representative of the population. Examples:

Hospital patients - use hospitals

Apartment dwellers - use apartment bldgs.

ee. Convenience Sampling

The subjects are convenient

Examples: Malls
 Street corners

Levels of Measurement

The Levels of Measurement

- a.) Nominal
- b.) Ordinal
- c.) Interval
- d.) Ratio

Qualities of Variables

Exhaustive: should include all possible answerable responses

Mutually exclusive: no respondent should be able to have two attributes simultaneously

Why is Level of Measurement Important?

Helps you decide what statistical analysis is appropriate on the values that were assigned.

Helps you decide how to interpret the data from that variable.

Nominal Measurement:

- a.) The values just "name" the attribute uniquely.
- b.) Does not imply any ordering of the cases.
- c.) For example, jersey numbers in football.
Even though player 32 has higher number than player 19, you can't say from the data that he's greater than or more than the other player.

Ordinal Measurement:

- a.) Attributes can be rank-ordered
- b.) Distances between attributes do not have any meaning.
For example: code Educational Attainment as 0= less than H.S.; 1 = some H.S.; 2 = H.S. degree; 3 = Some College; 4 = College degree; 5 = Post college.
Is the distance from 0 to 1 the same as 3 to 4?

Interval Measurement:

- a.) Distance between attributes has meaning
- b.) For example: Temperature (in Fahrenheit) Distance from 30-40 is the same as the distance from 70-80.
- c.) Note that ratios don't make any sense. 80 degrees is not twice as hot as 40 degrees.

Ratio Measurement:

- a.) Has an absolute zero that is meaningful.
- b.) Can construct a meaningful ratio (fraction).
- c.) For example: Number of clients in past six months.
- d.) It is meaningful to say that ". . .we had twice as many clients in this period as we did in the previous six months.